

# Shreeram Bhattarai

AI and Data specialist, CIMPA GmBH, Hamburg, Germany  
srbhattarai13@gmail.com — +49 (171) 820-8305 — linkedin: shbhattarai

## RESEARCH INTERESTS

---

Quantum Computation, Quantum Information, Quantum Algorithms and Machine Learning, Quantum AI, Quantum Simulation

## EDUCATION

---

**University of Bonn, Bonn, Germany** Master of Science in Computer Science July, 2022  
*Thesis Title: Anomaly Detection by Predicting Future Representations in Videos*

**Tribhuvan University, Kathmandu, Nepal** Bachelor of Science in Computer Science December, 2014

## PROFESSIONAL EXPERIENCE

---

**CIMPA GmbH** Hamburg, Germany  
*AI and Data Specialist* August, 2022 — Present

- Developing different LLM-based RAG System.
- Researching and developing GenAI solutions for different business use cases.
- Designing and implementing a GenAI Proof of Concept (POC) application leveraging Microsoft Azure cloud services.
- Developing models to forecast trends and behaviors using machine-learning techniques inside azure/AWS cloud.
- Gathering and preprocessing data and conducting exploratory data analysis.
- Collaborating with marketing, finance, and operations teams to integrate data insights into business strategies.

**ALSO Deutschland GmbH** Soest, Germany  
*Artificial Intelligent Developer* March, 2022 — August, 2022

- Was part of the research and development team for developing AI-based products.
- Involved in the complete process of product development including research, data collection, neural network design, training, and deployment of the model .

**Spreadfilms GmbH** Traunstein, Germany  
*Software Engineer* July, 2021 — October, 2021

- Design and develop software applications to meet the needs of a diverse user community.
- Provide ongoing support to other engineers, programmers, quality assurance personnel or users.
- Inspect program logic for errors in programming

## RESEARCH EXPERIENCE

---

**Max-Planck-Institut für Intelligente Systeme** Tübingen, Germany  
*Machine Learning Research Internship* September 2020 — July 2021

- Worked on Fiber-Optic Shape Sensing Using Neural Networks Operating on Multispecklegrams.
- Strengthened capability of reading scientific papers and understanding complex ideas.
- Designed and Implemented a recursive neural network (RNN), Complex neural network, and Convolution neural network ( CNN) for optical fiber data.
- Trained and evaluated the network performance with performance metrics.
- Collaborated with diverse team members and published our work in IEEE journal.

**Cognex Corporation** Jena, Germany  
*Deep Learning Research Internship* March, 2020 — May 2020

- Worked on Novelty detection on images.
- Researched different closely related papers on novelty detection and implemented them for company use cases.
- Designed different experiments and Trained CNN-based deep learning models.
- Summarized the finding by evaluating the trained model and well documented for presentation within the team.
- Developed a better understanding of Agile methodology by working with the team.
- Improved communication and research skills.
- Strengthened technical skills (e.g. Pytorch framework), python programming, Scrum.

## Fraunhofer Institute for Algorithms and Scientific Computing SCAI

*Student Research Assistant*

Bonn, Germany  
May, 2019 — February, 2020

- Worked on a Machine learning-based Car crash prediction project under the supervision of Prof. Dr. Jochen Garcke.
- Developed an understanding of complex research ideas by reviewing literature in deep learning methods used in future prediction of time-dependent data.
- Gained better insight into creating a data pipeline for deep learning by involving in the preprocessing of the simulation data and implementation of a Long short-term memory(LSTM) neural network.

## Computer Vision Group (Prof. Dr. Jürgen Gall)

*Student Research Assistant*

Bonn, Germany  
November, 2017 — May, 2019

- Involved in the creation, annotation, and cleaning process of raw data for deep learning architectures.
- Performed training and evaluation of different neural network architectures.
- Developed a web application to create a common platform for action recognition researchers.
- Worked closely with Ph.Ds and PostDoc and learned collaborative skills

## PUBLICATIONS

---

### IEEE Sensors Journal

- C. G. L. Cao, B. Javot, S. Bhattarai, K. Bierig, I. Oreshnikov and V. V. Volchkov, "Fiber-Optic Shape Sensing Using Neural Networks Operating on Multispecklegrams," in IEEE Sensors Journal, vol. 24, no. 17, pp. 27532-27540, 1 Sept. 1, 2024.

## SELF LEARNING

---

- MIT 8.04 Quantum Physics I, Spring 2016
- Foundations of Quantum Theory: Division of Theoretical Physics of Institute of Physics
- Quantum Physics Full Course — Quantum Mechanics Course: Brant Carlson, Associate Professor of Physics and Astronomy, Carthage College
- MIT 18.435J — Graduate Course, Quantum Computation

## SKILLS

---

- **Quantum Computation framework:** Qiskit, Cirq
- **Machine learning:** Linear regression, logistic regression, SVM, Random forest, Decision tree.
- **Deep learning Framework:** Jax, Pytorch, Keras, Tensorflow
- **Programming:** Python, Javascript, HTML, C++
- **Software Development Tools:** GIT, Docker, SCRUM, JIRA, Agile
- **Cloud Platform:** MS AZURE, AWS Cloud

## LANGUAGES

---

**English** - Advanced (Professional)

**German** - Intermediate

**Nepali** - Native

**Hindi** - Intermediate